THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The point which divides the join of $(0, 0)$ and $(6, 9)$ in the ratio 1:2 internally is: 1

(a) $(2,\ 3)$ **(b)** $(4,\ 6)$ **(c)** $(3,\ \frac{9}{2})$ **(d)** $(6,\ 9)$

2. If $\cos A = \frac{5}{13}$, then $\tan A =$ 1

(a) $\frac{12}{5}$ **(b)** $\frac{12}{13}$ **(c)** $\frac{5}{12}$ **(d)** $\frac{13}{5}$

3. A ladder 6 m long leans against a vertical wall and makes an angle of $60°$ with the level ground. The height reached on the wall is: 1

(a) $3\sqrt{3}$ m **(b)** $6\sqrt{3}$ m **(c)** $\frac{3\sqrt{3}}{2}$ m **(d)** 3 m

4. A point P is 34 cm from the centre O of a circle of radius 16 cm. The length of the tangent drawn from P to the circle is: 1

(a) 30 cm **(b)** 50 cm **(c)** 900 cm **(d)** 18 cm

5. A wheel has radius 21 cm. The number of complete revolutions it makes to cover 66 m is: 1

(a) 60 **(b)** 50 **(c)** 25 **(d)** 100

6. A capsule is a cylinder of radius 3.5 cm and length 10 cm (straight part) with a hemisphere of the same radius fixed at each end. The total surface area of the capsule is: 1

(a) 748 cm² **(b)** 297 cm² **(c)** 451 cm² **(d)** 374 cm²

7. For the data with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 with frequencies 4, 3, 2, 4, 8, 4 respectively, The median of the data is: 1

(a) 70 **(b)** 78.75 **(c)** 73.4 **(d)** 80

8. A coin is tossed three times. The probability of getting exactly two heads is: 1

(a) $\frac{1}{2}$ **(b)** $\frac{5}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{8}$

9. The product of a non-zero rational and an irrational number such as $2\sqrt{3}$ is: 1

(a) always an integer **(b)** always irrational

(c) sometimes rational **(d)** always rational

10. A quadratic polynomial whose zeroes are 3 and 4 is: 1

(a) $x^2 - 7x + 12$ **(b)** $x^2 - 7x - 12$ **(c)** $x^2 + 7x + 12$ **(d)** $x^2 + 7x - 12$

11. The solution of $2x + 3y = 12$, $5x + 4y = 23$ has $y =$ 1

(a) 3 **(b)** 5 **(c)** -2 **(d)** 2

12. If one root of $x^2 - 9x + k = 0$ is 2, then k is: 1

(a) 9 **(b)** 14 **(c)** 16 **(d)** 7

13. If $a_n = 3n + 2$, then the common difference is: 1

(a) 5 **(b)** 2 **(c)** 3 **(d)** 6

14. The length of the diagonal of a square of side 3 cm, squared, is: 1

(a) 9 **(b)** 36 **(c)** 6 **(d)** 18

15. The area of the triangle with vertices $(0, 0)$, $(4, 0)$ and $(0, 6)$ is: 1

(a) 5 **(b)** 24 **(c)** 12 **(d)** 10

16. The value of $\sin^2 30^\circ + \cos^2 30^\circ$ is: 1

(a) $\frac{1}{4}$ **(b)** 1 **(c)** $\frac{3}{2}$ **(d)** 2

17. A boy whose eyes are 2 m above the ground stands 10 m away from a vertical building. The angle of elevation of the top of the building from his eyes is 45°. The height of the building is: 1

(a) $12\sqrt{3}$ m **(b)** 13 m **(c)** 12 m **(d)** 10 m

18. A quadrilateral $ABCD$ is drawn about a circle so that all four sides touch the circle. $AB = 9$ cm, $BC = 11$ cm and $CD = 7$ cm. The length of AD is: 1

(a) 5 cm **(b)** 13 cm **(c)** 27 cm **(d)** 9 cm

19. The radii of two circles are 19 cm and 9 cm. The radius of the circle whose circumference equals the sum of their circumferences is: 1

(a) 28 cm **(b)** 14 cm **(c)** 30 cm **(d)** 56 cm

20. A solid is a cube of edge 14 cm surmounted by a hemisphere of diameter 14 cm (the hemisphere rests on a face of the cube). The total surface area of the solid is: 1

(a) 1176 cm² **(b)** 1022 cm² **(c)** 1330 cm² **(d)** 1484 cm²

SECTION — B (Very Short Answer Type) — 6 × 2 = 12

Answer in about 20–30 words each.

21. Find the distance between the points $(2, 3)$ and $(4, 1)$. 2

22. If $\tan A = \frac{4}{3}$, find $\sec A$. 2

23. A storm breaks a tree at a height of 6 m above the ground. The broken part bends over, its top touches the ground and the broken part makes an angle of 30° with the ground. Find the distance from the foot of the tree to the point where its top touches the ground. 2

24. Tangents QA and QB are drawn from an external point Q to a circle, and $QA = (3x + 4)$ cm and $QB = (2x + 9)$ cm. Find the length of QA. 2

25. The minute hand of a clock is 14 cm long. Find the distance moved by its tip in 15 minutes. 2

26. A solid metallic sphere of radius 6 cm is melted and drawn into a wire of circular cross-section of radius 2 cm. Find the length of the wire. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. The angle of elevation of the top of a vertical tower from a point on level ground is 30°. After walking 20 m towards the foot of the tower along a straight line, the angle of elevation becomes 60°. Find the height of the tower. 3

28. The incircle of $\triangle ABC$ touches BC, CA and AB at D, E and F. $AB = 12$ cm, $BC = 7$ cm and $CA = 9$ cm. Find the lengths AF, BD and CE (in this order). 3

29. A chord of a circle of radius 7 cm subtends a right angle at the centre. Find the area of the minor segment. 3

30. A toy is a cone of base radius 21 cm and height 10 cm mounted on a hemisphere of the same radius. Find the volume of the toy. 3

31. The marks obtained by students in a test are given below. 3

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Number of students	20	22	20	17	6	15

Find the mean.

OR

Two dice are thrown simultaneously. Find the probability that the product is a perfect square. 3

32. Show that $3 + 2\sqrt{5}$ is irrational, given that $\sqrt{5}$ is irrational. 3

OR

If 2 is a zero of the polynomial $x^2 + kx + 6$, find k and the other zero. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. 5

34. A chord of a circle of radius 14 cm subtends an angle of 60° at the centre. (Use $\pi = \frac{22}{7}$ and $\sqrt{3} = 1.73$; answer to two decimal places.) Find the area of the minor segment. 5

35. Water from a full cylindrical tank of radius 7 cm and height 10 cm is poured into an empty rectangular tank of base 11 cm by 7 cm. Find the height of water in the rectangular tank. 5

36. The mean of a distribution with classes 40–50, 50–60, 60–70, 70–80, 80–90, 90–100 and frequencies 4, 3, 2, p, 8, 4 respectively is 73.4. Find the value of p. 5

OR

Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 8,
(ii) both dice show the same number (a doublet),
(iii) the product of the numbers is 12.

37. A sweet seller has 420 kg of kaju barfi and 130 kg of badam barfi. She packs them into trays of the same weight, each tray holding one kind only. Find the maximum weight of a tray and the number of trays. 5

OR

Given that 1 and 2 are two zeroes of $x^3 - 8x^2 + 17x - 10$, find the third zero. 5

38. The taxi charges consist of a fixed charge plus a charge per km. For 10 km the fare is ₹170 and for 15 km it is ₹240. Find the fixed charge, the charge per km and the fare for 25 km. 5

OR

The length of a rectangular plot exceeds its breadth by 7 m and its area is 330 m². Find its dimensions. 5

END OF QUESTION PAPER — Total: 80 Marks